TOPSTECHNOLOGIES
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RADIUS & TACACS+ : TACACS+ & RADIUS Features, RADIUS Authentication Process, TACACS+ Use Cases, RADIUS vs. TACACS+

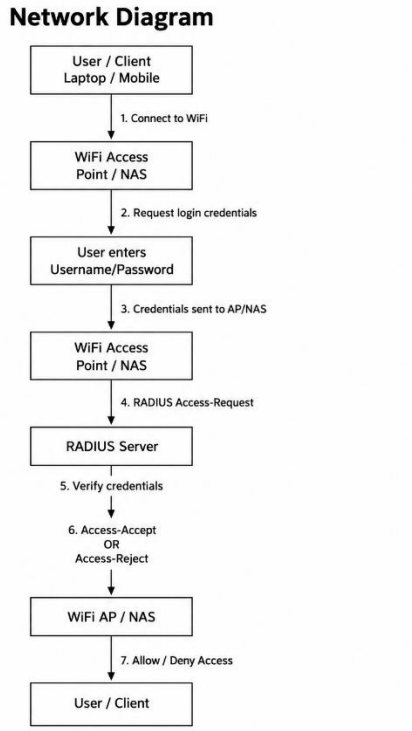
1. Create a comparison table listing at least 4 key features of RADIUS and TACACS+ side by side, highlighting which protocol supports each feature.

ANS :

Feature	**RADIUS**	**TACACS+**
Full Form	Remote Authentication Dial-In User Service	Terminal Access Controller Access-Control System Plus
Transport Protocol	UDP	TCP
Default Ports	UDP 1812 Authentication, UDP 1813 Accounting	TCP 49
Authentication	Supported	Supported
Authorization	Supported, but authentication and authorization are closely integrated	Supported separately from authentication
Accounting	Supported	Supported
Packet Encryption	Encrypts mainly the user's password	Encrypts almost the entire packet payload
Command-by-command Authorization	Limited / not normally used	Strong support
Reliability	Uses connectionless UDP	Uses reliable connection-oriented TCP
Typical Usage	Wi-Fi, VPN, 802.1X, network access	Router, switch, firewall and administrator access
Vendor Support	Multi-vendor standard and widely supported	Widely used especially for network-device administration
AAA Separation	Authentication and authorization are generally combined	Authentication, authorization and accounting can be handled independently

2. Draw a step-by-step flow diagram showing the RADIUS authentication process for a user logging in to a public WiFi hotspot, labeling each communication step between the client, network device, and RADIUS server.

ANS :



Step-by-Step RADIUS Authentication

Step 1 – User connects to WiFi

The user selects a public or college WiFi network.

Example:

College-WiFi

The user's laptop or mobile device connects to the wireless access point.

Step 2 – Network asks for credentials

The network may ask the user to enter authentication information such as:

Username: student101

Password: ********

These credentials identify the user.

Step 3 – Network device receives credentials

The WiFi Access Point or **NAS (Network Access Server)** receives the authentication information.

The NAS acts as the **RADIUS client**.

The user normally does not communicate directly with the RADIUS server.

Step 4 – Access-Request

The NAS sends a **RADIUS Access-Request** message to the RADIUS server.

It can contain information such as:

Username

Password/authentication data

NAS IP Address

Device information

Connection information

Example:

WiFi AP
|
| RADIUS Access-Request
|
v
RADIUS Server

Step 5 – RADIUS server verifies the user

The RADIUS server checks the supplied credentials against an identity database.

For example:

Active Directory

LDAP

Local Database

Student Database

If the username and authentication information are valid, authentication succeeds.

Step 6 – Server sends the response

The RADIUS server sends one of the appropriate responses.

Access-Accept

The user's authentication is successful.

Access-Reject

The user's authentication has failed.

RADIUS can also use:

Access-Challenge

when additional authentication information is required.

Step 7 – Access is allowed or denied

If the NAS receives:

Access-Accept

the user is given network access.

If it receives:

Access-Reject

3. List 3 real-world use cases where TACACS+ is preferred over RADIUS, and briefly explain why TACACS+ is better suited for each scenario.

ANS:

1. Network Administrator Access to Routers

Scenario

A company has many Cisco routers. Network administrators use SSH to configure them.

Administrator
|
| SSH
v
Cisco Router
|
| TACACS+
v
AAA Server

Why TACACS+ is preferred

TACACS+ provides detailed authorization for administrative commands.

For example, one administrator may be allowed to use:

show running-config

show ip interface brief

ping

but may be prevented from using critical commands.

Because TACACS+ supports detailed authorization, it is well suited for administrative device access.

2. Managing Enterprise Network Switches

Scenario

A university has hundreds of managed switches across different buildings.

Different IT staff members have different responsibilities.

For example:

Junior Technician

Network Engineer

Senior Administrator

Why TACACS+ is preferred

TACACS+ separates:

Authentication

Authorization

Accounting

Therefore, different privilege levels and permissions can be assigned to different administrators.

Example:

Junior Technician
↓
Show commands only

Network Engineer
↓
Configuration commands

Senior Administrator
↓
Full administrative access

This provides better administrative control.

3. Firewall Administration

Scenario

A company has critical firewalls protecting its network.

Only authorized security administrators should be allowed to make firewall changes.

Security Administrator

|

| SSH / HTTPS

v

Firewall

|

| TACACS+

v

AAA Server

Why TACACS+ is preferred

TACACS+ can provide detailed command authorization and accounting.

The organization can keep records of administrative actions such as:

- Who logged in
- When they logged in
- Which commands were executed
- What privilege level was used
- When the session ended

This is useful for:

- Security auditing
- Troubleshooting
- Compliance
- Change tracking

4. Given a scenario where a user tries to log into a network device but fails authentication, describe how RADIUS and TACACS+ would handle the logging and response differently. Hint Focus on how each protocol manages accounting and command authorization.

ANS:

Suppose a user tries to access a network device with incorrect login credentials.

Example:

Username: admin

Password: wrong123

RADIUS Process

User
|
| Login Attempt
v
Network Device
|
| Access-Request
v
RADIUS Server
|
| Check Credentials
|
| Authentication Failed
v
Access-Reject
|
v
Network Device
|
v
Access Denied

The RADIUS server records the failed authentication attempt depending on the server's logging configuration.

Possible information recorded includes:

Username

Timestamp

NAS/device IP

Authentication result

Reason for rejection

The network device receives:

Access-Reject

and refuses access.

RADIUS accounting is commonly focused on network access sessions.

TACACS+ Process

Administrator
|
| Login Attempt
v
Network Device
|
| Authentication Request
v
TACACS+ Server
|
| Verify Credentials
|
| Authentication Failed
v
Authentication Failure
|
v
Network Device
|
v
Access Denied

The TACACS+ server can maintain detailed records related to administrator access.

If authentication succeeds, TACACS+ can separately determine which commands that administrator may execute.

For example:

User authenticated successfully
|
v
show running-config
|
v

Authorized

But another command could be denied:

configure terminal
|
v
Authorization Check
|
v
Denied

RADIUS vs TACACS+ During Failed Login

Area	**RADIUS**	**TACACS+**
Authentication Failure	Access-Reject	Authentication failure response
Logging	Can log authentication attempts and session/accounting information	Can provide detailed administrator AAA records
Authorization	Less granular for network-device command control	Detailed command authorization
Command Tracking	Not its primary strength	Strong command-level control/accounting capabilities
Typical Failed Login	WiFi/VPN/network user	Router/switch/firewall administrator

Important Difference

RADIUS is generally optimized for:

"Should this user be allowed onto the network?"

TACACS+ is particularly useful for questions such as:

"Can this administrator log in?"

"Which privilege level should they receive?"

"Which administrative commands are they allowed to execute?"

"What administrative actions were performed?"

5.Use ChatGPT or another AI tool to generate a sample RADIUS configuration snippet for a Cisco router, then review the output and list any changes you would make to adapt it for a college campus WiFi setup.

ANS:

Below is an example Cisco IOS-style configuration. Exact syntax can vary slightly depending on the IOS/IOS XE version.

```
Router> enable
Router# configure terminal
Router(config)# aaa new-model
Router(config)# radius server COLLEGE-RADIUS
Router(config-radius-server)# address ipv4 192.168.10.10 auth-port 1812 acct-port 1813
Router(config-radius-server)# key CollegeRadius123
Router(config-radius-server)# exit
Router(config)# aaa authentication login default group radius local
Router(config)# aaa authorization exec default group radius local
Router(config)# aaa accounting exec default start-stop group radius
Router(config)# aaa accounting network default start-stop group radius
Router(config)# line vty 0 4
Router(config-line)# login authentication default
Router(config-line)# transport input ssh
Router(config-line)# exit
Router(config)# end
Router# write memory
```

Explanation of Configuration

1. Enable AAA

aaa new-model

Enables Authentication, Authorization and Accounting on the Cisco device.

2. Create the RADIUS server definition

radius server COLLEGE-RADIUS

Creates a RADIUS server profile called:

COLLEGE-RADIUS

3. Configure RADIUS server IP and ports

address ipv4 192.168.10.10 auth-port 1812 acct-port 1813

The RADIUS server IP is:

192.168.10.10

Authentication uses:

UDP 1812

Accounting uses:

UDP 1813

4. Configure the shared secret

key CollegeRadius123

The Cisco device and RADIUS server use a shared secret.

The same secret must be configured on both sides.

In a real environment, a stronger secret should be used.

5. Configure authentication

aaa authentication login default group radius local

The router first attempts authentication using RADIUS.

If the RADIUS service is unavailable, it can fall back to the router's local user database.

6. Configure authorization

aaa authorization exec default group radius local

RADIUS can provide authorization information for EXEC sessions.

7. Configure accounting

aaa accounting exec default start-stop group radius

This can record the beginning and end of EXEC sessions.

aaa accounting network default start-stop group radius

This can record network-related session activity where applicable.

8. Secure remote access

line vty 0 4

login authentication default

transport input ssh

This applies AAA authentication to remote VTY sessions and permits SSH rather than insecure Telnet access.

Changes for a College Campus WiFi Environment

For a real college campus, I would modify and expand the sample configuration.

1. Replace the Sample RADIUS Server Address

Instead of:

192.168.10.10

use the actual college RADIUS server address.

Example:

10.10.100.10

2. Use a Strong Shared Secret

Instead of:

CollegeRadius123

use a long, randomly generated shared secret and protect it from unauthorized disclosure.

3. Integrate the RADIUS Server with a Student Database

The college could connect RADIUS authentication to:

Microsoft Active Directory

LDAP

Student Information System

Faculty Database

Example:

Student WiFi Login
|
v
RADIUS
|
v
Active Directory

4. Use 802.1X / WPA2-Enterprise or WPA3-Enterprise

For campus WiFi, RADIUS would normally be used with enterprise wireless authentication.

Example architecture:

Student Laptop
|
| 802.1X
v
Wireless AP
|
v
Wireless Controller / NAS
|
| RADIUS
v
RADIUS Server
|
v
College Identity Database

This gives each student or staff member an individual login instead of relying only on one shared WiFi password.

5. Create Different Access Policies

Different users can be placed into different access groups.

Example:

User Type	Network Access
Students	Student VLAN
Faculty	Faculty VLAN
IT Staff	IT VLAN
Guests	Guest VLAN

Example:

RADIUS Server

|

+---- Student → VLAN 10

|

+---- Faculty → VLAN 20

|

+---- IT Staff → VLAN 30

|

+---- Guest → VLAN 40

6. Configure Accounting

RADIUS accounting should be enabled to record details such as:

Username

Login time

Logout time

NAS / access point

Session duration

Assigned IP address

This is useful for troubleshooting and security auditing.

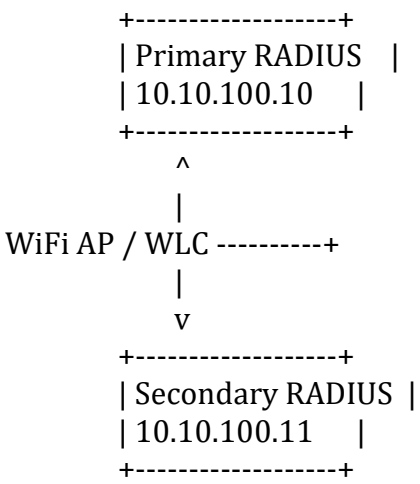
7. Deploy Backup RADIUS Servers

Instead of relying on only one server:

RADIUS Server 1

the college should ideally have redundancy.

Example:



If the primary server becomes unavailable, authentication can be directed to the secondary server according to the network design.